EFSP Auxiliary ESL Curriculum

Manufacturing English

Instructor guide for advanced ESL learners working in manufacturing

Audience: plant managers, production supervisors, quality engineers, maintenance teams, industrial engineers, safety leads, supply planners, and manufacturing-adjacent professionals

Focus: A manufacturing English curriculum for production meetings, line problems, lean improvement, defects, maintenance, safety, supplier quality, shift handoffs, and root-cause communication.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

Purpose and Course Logic

A manufacturing English curriculum for production meetings, line problems, lean improvement, defects, maintenance, safety, supplier quality, shift handoffs, and root-cause communication.

Core language challenge

Advanced learners do not only need vocabulary. They need the ability to ask which standard applies, who owns the decision, what evidence is sufficient, what risk is being accepted, and how to disagree without sounding vague, defensive, or reckless.

Each module trains a realistic workplace pressure point with role-specific terms, decision language, pushback practice, and a written output learners can adapt to their own work.

Course objectives

- Use manufacturing terminology accurately in meetings, written updates, handoffs, escalations, reviews, and client or stakeholder conversations.
- Turn vague requests into specific questions about evidence, owner, deadline, constraint, risk, and decision rights.
- Push back on unsafe, unsupported, noncompliant, unrealistic, or poorly scoped proposals while preserving professional trust.
- Handle realistic dialogues from the field, including conflict, uncertainty, documentation gaps, customer or stakeholder pressure, and cross-functional disagreement.
- Produce concise workplace outputs: briefing notes, escalation updates, meeting scripts, risk memos, decision records, and follow-up messages.

Instructor Module Plans

Module 1. Production Flow and Daily Management (90 minutes)

Report line status with throughput, downtime, and constraint language.

Learners should be able to

- Use these terms accurately: throughput, cycle time, constraint, downtime.
- Explain the workplace tension: The constraint may be material availability, changeover time, staffing, or equipment reliability.
- Respond professionally when a stakeholder says: Ask operators to work faster.
- Draft a usable daily production update with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

A line misses target for the third shift in a row.

Ask operators to work faster.

The constraint may be material availability, changeover time, staffing, or equipment reliability.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.

4. Output lab: draft and revise a daily production update.

Module 2. Lean, Waste, and Continuous Improvement (90 minutes)

Discuss waste reduction without blaming people.

Learners should be able to

- Use these terms accurately: kaizen, standard work, waste, value stream.
- Explain the workplace tension: Process design, layout, standard work, and visual management need review.
- Respond professionally when a stakeholder says: Tell employees to be more efficient.
- Draft a usable lean improvement proposal with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

A kaizen event identifies excess movement and waiting.

Tell employees to be more efficient.

Process design, layout, standard work, and visual management need review.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.
4. Output lab: draft and revise a lean improvement proposal.

Module 3. Defects, Scrap, and Rework (90 minutes)

Explain defect trends and containment actions clearly.

Learners should be able to

- Use these terms accurately: defect, scrap, rework, containment.
- Explain the workplace tension: Containment, defect mode, inspection plan, and customer impact must be defined.
- Respond professionally when a stakeholder says: Ship the acceptable units and watch the trend.
- Draft a usable defect containment brief with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

Scrap has increased after a tooling change.

Ship the acceptable units and watch the trend.

Containment, defect mode, inspection plan, and customer impact must be defined.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.

4. Output lab: draft and revise a defect containment brief.

Module 4. Root Cause and Corrective Action (90 minutes)

Move from symptom language to evidence-based cause analysis.

Learners should be able to

- Use these terms accurately: root cause, 5 Whys, fishbone, 8D.
- Explain the workplace tension: Root cause must be supported by data, not assumed from the visible failure.
- Respond professionally when a stakeholder says: Say the operator missed a step.
- Draft a usable 8D problem statement with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

A customer returns parts for fit issues.

Say the operator missed a step.

Root cause must be supported by data, not assumed from the visible failure.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.
4. Output lab: draft and revise a 8D problem statement.

Module 5. Maintenance, Reliability, and Changeover (90 minutes)

Discuss equipment risk and maintenance tradeoffs.

Learners should be able to

- Use these terms accurately: preventive maintenance, MTBF, changeover, spare parts.
- Explain the workplace tension: Unplanned downtime, safety risk, spare parts, and preventive maintenance need balancing.
- Respond professionally when a stakeholder says: Delay maintenance until the order is complete.
- Draft a usable maintenance risk escalation with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

A critical machine keeps failing during peak demand.

Delay maintenance until the order is complete.

Unplanned downtime, safety risk, spare parts, and preventive maintenance need balancing.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.

3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.
4. Output lab: draft and revise a maintenance risk escalation.

Module 6. EHS and Safety Communication (90 minutes)

Stop unsafe work with direct but professional language.

Learners should be able to

- Use these terms accurately: EHS, lockout/tagout, near miss, stop work.
- Explain the workplace tension: Safety controls, lockout/tagout, and incident risk override schedule pressure.
- Respond professionally when a stakeholder says: Finish the run and fix the guard later.
- Draft a usable safety stop-work script with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

A supervisor sees bypassed guarding during a rush order.

Finish the run and fix the guard later.

Safety controls, lockout/tagout, and incident risk override schedule pressure.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.
4. Output lab: draft and revise a safety stop-work script.

Module 7. Supplier Quality and Incoming Materials (90 minutes)

Communicate supplier problems with evidence and business impact.

Learners should be able to

- Use these terms accurately: supplier quality, incoming inspection, specification, deviation.
- Explain the workplace tension: Specification, deviation approval, alternate supply, and customer risk must be assessed.
- Respond professionally when a stakeholder says: Use it because the supplier says it is fine.
- Draft a usable supplier deviation request with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

Incoming material fails inspection before a major build.

Use it because the supplier says it is fine.

Specification, deviation approval, alternate supply, and customer risk must be assessed.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.

2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.
4. Output lab: draft and revise a supplier deviation request.

Module 8. Shift Handoffs and Escalation (90 minutes)

Create clean shift-to-shift continuity.

Learners should be able to

- Use these terms accurately: handoff, andon, escalation, owner.
- Explain the workplace tension: Handoff needs status, actions taken, risk, owner, and escalation path.
- Respond professionally when a stakeholder says: Let day shift figure it out.
- Draft a usable shift handoff note with facts, caveats, owner, and next step.

Customized scenario

Workplace pressure

The night shift leaves an unresolved process alarm.

Let day shift figure it out.

Handoff needs status, actions taken, risk, owner, and escalation path.

Classroom sequence

1. Terminology drill: define each term, then use it in one sentence from the learner's own role.
2. Risk map: identify the stakeholder, the decision, the evidence gap, the operating constraint, and the cost of being wrong.
3. Pushback ladder: move from clarifying question to evidence-based objection to consequence to decision request.
4. Output lab: draft and revise a shift handoff note.

Nomenclature and Jargon

These are classroom working definitions. Learners should adapt wording to their organization's policies, systems, and local regulatory environment.

Production Flow and Daily Management

Term	Working meaning
throughput	Amount of work, patients, goods, cases, or transactions completed in a period of time.
cycle time	Working manufacturing term used in production flow and daily management; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
constraint	Working manufacturing term used in production flow and daily management; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
downtime	Working manufacturing term used in production flow and daily management; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

Lean, Waste, and Continuous Improvement

Term	Working meaning
kaizen	Working manufacturing term used in lean, waste, and continuous improvement; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
standard work	Working manufacturing term used in lean, waste, and continuous improvement; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
waste	Working manufacturing term used in lean, waste, and continuous improvement; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
value stream	Working manufacturing term used in lean, waste, and continuous improvement; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

Defects, Scrap, and Rework

Term	Working meaning
defect	Failure, flaw, nonconformance, or error that prevents an output from meeting requirements.
scrap	Working manufacturing term used in defects, scrap, and rework; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
rework	Working manufacturing term used in defects, scrap, and rework; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
containment	Working manufacturing term used in defects, scrap, and rework; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

Root Cause and Corrective Action

Term	Working meaning
root cause	Underlying reason a problem occurred, not merely the visible symptom.
5 Whys	Working manufacturing term used in root cause and corrective action; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
fishbone	Working manufacturing term used in root cause and corrective action; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
8D	Working manufacturing term used in root cause and corrective action; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

Maintenance, Reliability, and Changeover

Term	Working meaning
preventive maintenance	Working manufacturing term used in maintenance, reliability, and changeover; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
MTBF	Working manufacturing term used in maintenance, reliability, and changeover; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
changeover	Working manufacturing term used in maintenance, reliability, and changeover; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
spare parts	Working manufacturing term used in maintenance, reliability, and changeover; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

EHS and Safety Communication

Term	Working meaning
EHS	Working manufacturing term used in ehs and safety communication; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
lockout/tagout	Working manufacturing term used in ehs and safety communication; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
near miss	Working manufacturing term used in ehs and safety communication; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
stop work	Working manufacturing term used in ehs and safety communication; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

Supplier Quality and Incoming Materials

Term	Working meaning
supplier quality	Working manufacturing term used in supplier quality and incoming materials; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
incoming inspection	Working manufacturing term used in supplier quality and incoming materials; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
specification	Working manufacturing term used in supplier quality and incoming materials; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
deviation	Working manufacturing term used in supplier quality and incoming materials; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.

Shift Handoffs and Escalation

Term	Working meaning
handoff	Transfer of work, responsibility, information, or risk from one person or team to another.
andon	Working manufacturing term used in shift handoffs and escalation; define the owner, evidence source, governing document, risk, and decision impact before using it in a meeting.
escalation	Raising an issue to a higher authority or different function because risk, urgency, or decision rights require it.
owner	Named person or role accountable for a decision, action, deliverable, or risk.

Industry-Specific Meeting Moves

Situation	Useful language
Production Flow and Daily Management	Before we commit, I want to confirm throughput, cycle time, the owner, and the evidence behind the decision. If the constraint may be material availability, changeover time, staffing, or equipment reliability., I recommend we document the risk and agree on the next step.
Lean, Waste, and Continuous Improvement	Before we commit, I want to confirm kaizen, standard work, the owner, and the evidence behind the decision. If process design, layout, standard work, and visual management need review., I recommend we document the risk and agree on the next step.
Defects, Scrap, and Rework	Before we commit, I want to confirm defect, scrap, the owner, and the evidence behind the decision. If containment, defect mode, inspection plan, and customer impact must be defined., I recommend we document the risk and agree on the next step.
Root Cause and Corrective Action	Before we commit, I want to confirm root cause, 5 Whys, the owner, and the evidence behind the decision. If root cause must be supported by data, not assumed from the visible failure., I recommend we document the risk and agree on the next step.

Situation	Useful language
Maintenance, Reliability, and Changeover	Before we commit, I want to confirm preventive maintenance, MTBF, the owner, and the evidence behind the decision. If unplanned downtime, safety risk, spare parts, and preventive maintenance need balancing., I recommend we document the risk and agree on the next step.
EHS and Safety Communication	Before we commit, I want to confirm EHS, lockout/tagout, the owner, and the evidence behind the decision. If safety controls, lockout/tagout, and incident risk override schedule pressure., I recommend we document the risk and agree on the next step.
Supplier Quality and Incoming Materials	Before we commit, I want to confirm supplier quality, incoming inspection, the owner, and the evidence behind the decision. If specification, deviation approval, alternate supply, and customer risk must be assessed., I recommend we document the risk and agree on the next step.
Shift Handoffs and Escalation	Before we commit, I want to confirm handoff, andon, the owner, and the evidence behind the decision. If handoff needs status, actions taken, risk, owner, and escalation path., I recommend we document the risk and agree on the next step.

High-pressure pushback frames

- I understand the urgency. The risk is that we move faster than the evidence or process supports.
- I am not blocking the goal. I am naming the condition we need before the decision is safe and credible.
- If we accept this risk, we should name the owner, document the assumption, and define the trigger for escalation.
- That may be possible, but not under the current scope, timeline, or approval path.
- Let's separate what we know, what we assume, and what still needs confirmation.

Assessment and Coaching

Performance rubric

Skill	Developing	Proficient	Strong
Terminology	Recognizes terms but uses them loosely.	Uses field terms accurately in context.	Defines terms, connects them to evidence, and explains decision impact.
Pushback	Disagrees vaguely or avoids disagreement.	Names concern with evidence and next step.	Balances urgency, relationship, risk, owner, and decision rights.
Scenario judgment	Focuses on one stakeholder's preference.	Identifies constraint, risk, and process.	Guides the group toward a documented, realistic decision.
Written output	Writes general summaries.	Produces clear notes with facts and owner.	Creates concise, decision-ready workplace communication.

Source orientation

- Company quality manuals and SOPs.
- OSHA or local safety requirements.
- Customer specifications and supplier quality agreements.
- The learner's own company policies, SOPs, contracts, systems, templates, and approved communication standards.